

Worksheet 4A: Derivatives and Integrals — Inverse Operations

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

***What This Worksheet Is About** Derivatives and integrals are **inverses** of each other, just like multiplication and division, or squaring and square roots. - **Derivative:** Given a function, find its rate of change - **Integral:** Given a rate of change, find the original function But there's a twist: When we "undo" a derivative, we don't get back exactly one function — we get a whole **family** of functions!*

Part A: The Power Rule — Both Directions (20 problems)

Differentiation: Bring down the power, reduce by 1

$$\frac{d}{dx}x^n = n \cdot x^{n-1}$$

Integration: Increase the power by 1, divide by new power

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Notice they're inverses! - Derivative: exponent comes **down** and **decreases** - Integral: exponent goes **up** and we **divide** by it

Find both the derivative and the integral of each function:

1. $f(x) = x^4$ - $f'(x) =$ _____ - $\int x^4 dx =$ _____
2. $g(t) = t^5$ - $g'(t) =$ _____ - $\int t^5 dt =$ _____
3. $h(p) = p^3$ - $h'(p) =$ _____ - $\int p^3 dp =$ _____
4. $y = x^7$
5. $f(m) = m^2$
6. $g(r) = r^{10}$
7. $h(k) = k^1 = k$
8. $y = x^{100}$ **Negative and Fractional Exponents:**
9. $f(x) = x^{-2}$ - $f'(x) =$ _____ - $\int x^{-2} dx =$ _____

10. $g(t) = t^{-3}$
 11. $h(x) = x^{1/2} = \sqrt{x}$ - $h'(x) =$ _____ - $\int x^{1/2} dx =$ _____
 12. $y = x^{1/3}$
 13. $f(p) = p^{-1/2}$
 14. $g(n) = n^{3/2}$ **The Special Case: What happens when $n = -1$?**
 15. Try to compute $\int x^{-1} dx$ using the formula $\frac{x^{n+1}}{n+1}$. What goes wrong?
 16. The integral of $x^{-1} = \frac{1}{x}$ is actually $\ln|x| + C$. We'll explore why in Worksheet 4D!
- Checking Your Work — Derivatives and Integrals Undo Each Other:**
17. If $\int x^3 dx = \frac{x^4}{4} + C$, verify by differentiating: $\frac{d}{dx} \left(\frac{x^4}{4} + C \right) =$ _____
 18. If $\frac{d}{dx}(x^5) = 5x^4$, verify by integrating: $\int 5x^4 dx =$ _____
 19. Differentiate, then integrate: Start with $f(x) = x^6$
 - $f'(x) =$ _____
 - $\int f'(x) dx =$ _____
 - Do you get back x^6 ? (Plus what?)
 20. Integrate, then differentiate: Start with $g(x) = x^4$
 - $\int g(x) dx =$ _____
 - $\frac{d}{dx} [\int g(x) dx] =$ _____
 - Do you get back x^4 ?

Part B: Why "+C"? — The Family of Antiderivatives (14 problems)

Consider these three functions: - $f(x) = x^2$ - $g(x) = x^2 + 5$ - $h(x) = x^2 - 17$

21. Find the derivative of each:
 - $f'(x) =$ _____
 - $g'(x) =$ _____
 - $h'(x) =$ _____
22. What do you notice? Why does this happen?
23. The functions x^2 , x^2+5 , and x^2-17 are all different, but they have the _____ derivative.
24. Graphically: Sketch $y = x^2$, $y = x^2 + 3$, and $y = x^2 - 2$ on the same axes. - How are these parabolas related? - At any given x-value, do they have the same slope? **The Key Insight:**

When we integrate $2x$, we get $x^2 + C$ where C is **any constant**. This represents a whole **family** of functions — infinitely many parabolas, all with the same shape but shifted up or down.

25. Write three different functions whose derivative is $3x^2$.
26. Write three different functions whose derivative is $5t^4$.
27. Write the general antiderivative (with $+C$) of $4p^3$.
28. If $F'(x) = 6x^5$, what is $F(x)$? **Finding a Specific Function — Initial Conditions:**
If we know $F'(x) = 2x$ AND $F(0) = 7$, we can find the specific function: $F(x) = x^2 + C$ $F(0) = 0^2 + C = 7$, so $C = 7$ Therefore $F(x) = x^2 + 7$
29. Find $F(x)$ if $F'(x) = 3x^2$ and $F(0) = 5$.
30. Find $G(t)$ if $G'(t) = 4t^3$ and $G(1) = 10$.
31. Find $H(p)$ if $H'(p) = 2p + 1$ and $H(2) = 0$.
32. A ball's velocity is $v(t) = 20 - 10t$ m/s. Its position is $s(t) = \int v(t) dt$. - (a) Find $s(t)$ with the constant C . - (b) If the ball starts at height $s(0) = 5$ meters, find C . - (c) Write the complete position function.
33. An object's acceleration is $a(t) = 6$ m/s². Starting from rest ($v(0) = 0$): - (a) Find velocity: $v(t) = \int a(t) dt$ - (b) If position $s(0) = 10$, find $s(t) = \int v(t) dt$.
34. Explain in your own words why integration has "+C" but differentiation doesn't.

Part C: Constants and Sums — Both Directions (16 problems)

Derivative Rules:

- $\frac{d}{dx}(c) = 0$ (constant rule) - $\frac{d}{dx}[cf(x)] = c \cdot f'(x)$ (constant multiple) - $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$ (sum rule)

Integral Rules (the reverses):

- $\int 0 dx = C$ (integrating zero gives a constant) - $\int c \cdot f(x) dx = c \cdot \int f(x) dx$ (constant multiple) - $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$ (sum rule)

Also: $\int c dx = cx + C$ (integrating a constant)

Find both the derivative and the integral:

35. $f(x) = 5x^3$ - $f'(x) =$ _____ - $\int 5x^3 dx =$ _____

36. $g(t) = -2t^4$

37. $h(p) = \frac{1}{2}p^6$
38. $y = 7$ (a constant function) - $\frac{dy}{dx} = \underline{\hspace{2cm}}$ - $\int 7 dx = \underline{\hspace{2cm}}$
39. $f(m) = -3$ **Sums and Differences:**
40. $f(x) = x^3 + x^2$ - $f'(x) = \underline{\hspace{2cm}}$ - $\int (x^3 + x^2) dx = \underline{\hspace{2cm}}$
41. $g(t) = t^4 - t^2$
42. $h(r) = 3r^2 + 2r - 5$
43. $y = 4x^3 - 6x^2 + 2x - 1$
44. $f(k) = k^5 - 3k^3 + 2k$
45. $g(n) = 2n^4 + n^3 - 5n^2 + 4n - 7$ **Verify by reversing:**
46. Check problem 40: Differentiate your integral. Do you get back $x^3 + x^2$?
47. Check problem 43: Integrate your derivative. Do you get back $4x^3 - 6x^2 + 2x - 1$ (plus C)? **Rewriting Before Integrating:**
48. Find $\int \frac{3}{x^2} dx$ *Hint: Rewrite as $3x^{-2}$ first*
49. Find $\int \frac{5}{\sqrt{t}} dt$
50. Find $\int \left(x + \frac{1}{x^2}\right) dx$

Part D: Polynomials — Complete Practice (16 problems)

Find the derivative:

51. $f(x) = 2x^5 - 3x^4 + x^3 - 7x + 2$
52. $g(t) = \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2}$
53. $h(p) = (p + 2)(p - 3)$ (*expand first*)
54. $y = (x - 1)^2$ **Find the integral:**
55. $\int (3x^2 - 4x + 5) dx$
56. $\int (t^4 - 2t^3 + t - 3) dt$
57. $\int (6m^5 - 4m^3 + 2m) dm$
58. $\int (u^2 + 1)^2 du$ (*expand first!*) **Mixed — Derivative or Integral as Specified:**
59. $\frac{d}{dx}(x^4 - 3x^2 + 2)$
60. $\int (4r^3 - 6r) dr$
61. $\frac{d}{dt}(t^6 + 2t^4 - t)$
62. $\int (5k^4 - 3k^2 + 1) dk$ **Round-Trip Problems:**

63. Start with $f(x) = x^5$.
- Differentiate: $f'(x) =$
 - Integrate your answer: $\int f'(x) dx =$
 - Is this $f(x)$? Explain.
64. Start with $g(t) = t^3 - 4t$.
- Integrate: $\int g(t) dt =$
 - Differentiate your answer.
 - Is this $g(t)$? Explain.
65. Why does integrating then differentiating give you back exactly what you started with, but differentiating then integrating gives you the original plus a constant?
66. Complete this analogy: Squaring and square roots are inverses: $(\sqrt{x})^2 = x$ and $\sqrt{x^2} = |x|$ (almost x , but not quite for negatives) Similarly: $\frac{d}{dx} [\int f(x) dx] =$ _____ and $\int \frac{d}{dx} [f(x)] dx =$ _____

Part E: Building Intuition — What's Really Happening (10 problems)

67. The derivative of position is velocity. The integral of velocity is _____.
68. The derivative of velocity is acceleration. The integral of acceleration is _____.
69. If you know an object's acceleration at all times, can you determine its exact position? What additional information do you need?
70. Match the physical quantities:

Function	Derivative	Integral
Position		
Velocity		
Acceleration		

71. A car's velocity is $v(t) = 3t^2$ m/s. - (a) What is the acceleration $a(t) = v'(t)$? - (b) What is the position $s(t) = \int v(t) dt$? - (c) If $s(0) = 0$, what is the specific position function? - (d) How far has the car traveled from $t = 0$ to $t = 2$?
72. An object's position is $s(t) = t^3 - 6t^2 + 9t$. - (a) Find velocity $v(t) = s'(t)$. - (b) Find acceleration $a(t) = v'(t)$. - (c) When is velocity zero? - (d) When is acceleration zero?
73. Revenue $R(q)$ depends on quantity sold q . - **Marginal revenue** is $R'(q)$ — the extra revenue from one more unit. - If you know marginal revenue, $\int R'(q) dq$ gives you _____.
74. Cost $C(q)$ depends on quantity produced. - **Marginal cost** is $C'(q)$. - If marginal cost is $MC(q) = 6q + 10$, find the cost function $C(q)$. - If fixed costs are $C(0) = 500$, find the specific cost function.

75. Population growth rate is $P'(t) = 1000 + 50t$ people per year. - (a) Find $P(t)$, the population function. - (b) If the population at $t = 0$ is 50,000, find the specific function. - (c) What is the population after 10 years?
76. Water flows into a tank at rate $r(t) = 100 - 2t$ liters per minute. - (a) The amount of water $W(t) = \int r(t) dt$. Find it. - (b) If the tank starts empty, $W(0) = 0$. Find the specific function. - (c) How much water is in the tank after 10 minutes? - (d) When does the tank stop filling (when $r(t) = 0$)?

Part F: The Fundamental Theorem Preview (6 problems)

The **Fundamental Theorem of Calculus** connects derivatives and integrals precisely:

$$\int_a^b f(x) dx = F(b) - F(a)$$

where $F'(x) = f(x)$.

This says: To find the integral from a to b , find ANY antiderivative F , then compute $F(b) - F(a)$.

(The $+C$ cancels out!)

Evaluate these definite integrals:

77. $\int_0^2 x^2 dx$ - Antiderivative: $F(x) = \frac{x^3}{3}$ - $F(2) - F(0) = \frac{8}{3} - 0 =$ _____

78. $\int_1^3 2x dx$

79. $\int_0^4 (3t^2 - 2t) dt$

80. $\int_{-1}^1 x^3 dx$ *Hint: What symmetry do you notice?*

81. $\int_1^4 \sqrt{x} dx$

82. $\int_0^3 (x^2 + 1) dx$